Khac-Long Nguyen

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Education

International School, Vietnam National University, Hanoi (VNUIS) — Hanoi, Vietnam
BEng. Automation and Informatics — *08/2021 - 07/2026*

- QS WUR Ranking 386 for Engineering and Technology (Source)
- GPA: 3.71/4.0 (Rank #3 of the Cohort)
- Transcripts: Click here

National University of Singapore (NUS) — Singapore
Exchange Semester I, AY25/26 — *08/2025 - 12/2025*

- QS WUR Ranking 12 for Engineering and Technology (Source)
- Courseworks: Robot Mechanics and Control (ME4245), Electric Drives and Control (EE4502), Calculus (MA2002).

Research Experience

AIM-HI Institute, National Chung Cheng University — Chiayi, Taiwan
Research Intern — *05/2025 - 07/2025*

- **Project:** Smart Sensing Technology for Smart Manufacturing.
  - Developed a data-driven approach to estimate dynamic cutting force model in milling process by combining nonlinear system analysis with qualitative simulation.
  - Processed and trained 12 datasets of the cutting forces, each containing multi-channel sensor signals (three-axis dynamometer, spindle speed, feed rate, and depth of cut).
  - Implemented visualization and error analysis of predicted vs. actual measured cutting forces using Python.
  - Results: Prediction accuracy of 60-70% compared to experimental ground-truth cutting force measurements.
  - **Supervisor:** Prof. Her-Terng Yau

Remote Research — *08/2025-Current*

- **Project:** Prediction of cutting force model using AI and data-driven methods
  - Implementing 03 machine learning models and 01 data-driven model to predict cutting force.
  - Training 27 experienced datasets collected by Taguchi method.
  - **Supervisor:** Prof. Her-Terng Yau

Intelligent Control and Artificial Intelligence Lab, VNUIS — Hanoi, Vietnam
Student Research — *09/2022 - Current*

- **Project:** Implementation Model Predictive Control for Machining System (Current)
  - Reviewing more than 30 papers in control of machining system and cutting force prediction.
  - Combining machine learning and MPC model to control machining system.
  - **Supervisor:** Assoc. Prof. Nguyen Nhu Tung
- **Project:** Implementation Adaptive Control for 3-DOF Parallel Robot (2024)
  - Designed mechanical model and physical prototype of 3-DOF delta parallel robot.
  - Applied adaptive fuzzy logic combined with backstepping control for robust trajectory tracking under uncertainties.
  - Simulation Results (Matlab/Simulink):
    - Reduced steady-state deviation at J2 from 1.2 rad (Backstepping) to nearly 0 rad (Backstepping + Fuzzy).
    - Settling time improved from 0.3 s to 0.15–0.2 s.
    - Coordinate tracking (x, y, z) stabilized with maximum deviation 0.35 rad.
  - Results: Published in Lecture Notes in Networks and Systems, Springer, Q4.
  - **Supervisor:** Dr. Le Xuan Hai
- **Project:** Detection of Personal Protective Equipment (PPE) in Construction Sites (2023)
  - Built a custom dataset of 2,632 real construction site images across 5 PPE classes (helmet, vest, gloves, boots, person)
  - Trained and evaluated YOLOv5s with 6 activation functions, SiLU achieved the best performance.
  - Results: Precision 84.8%, Recall 80.2%, mAP@0.5 83.9%, mAP@0.5–0.95 43.6%.
  - **Supervisor:** Dr. Kim Dinh Thai

Teaching Experience

Faculty of Engineering and Technology, VNUIS Hanoi, Vietnam
Teaching Assistant *09/2023 - 12/2024*

- Modeling and Simulation of Control Systems (Semester 1, AY24/25)
  - Assisted in teaching undergraduate course focused on modeling, analysis, and simulation of dynamic systems.
  - Supported lecturer in preparing tutorials, lab manuals, and course materials.
  - Mentored student groups on applying simulation tools for robotics, mechatronics and process control applications by using MATLAB/Simulink.
  - **Supervisor:** Dr. Le Xuan Hai
- PLC Programming (Semester 1, AY24/25)
  - Assisted in preparing technical manual documents in Siemens S7-1200 and TIA Portal.
  - Mentored 01 team programming 4-floor elevator control logic project and 01 team programming pneumatic valve and signal control project.
  - Assisted with troubleshooting, hardware interfacing, and simulation.
  - **Supervisor:** MSc. Pham Hai Yen
- Microprocessors and Microcontrollers (Semester 1, AY23/24)
  - Assisted in teaching undergraduate course covering embedded C, and microcontroller interfacing (Arduino, STM32).
  - Mentored students on debugging techniques using Keil uVision, Arduino IDE, and STM32CubeIDE.
  - Supported lecturer in preparing tutorials, lab manuals, and course materials.
  - **Supervisor:** Dr. Le Xuan Hai

Hung STEM Robotics Lab Hanoi, Vietnam
Collaborator *07/2022 - 12/2022*

- **Project**: World Robotics Olympiad 2022 Vietnam
  - Mentored in a team to participate in World Robotics Olympiad 2022 Vietnam, assisted the team in building up new LEGO modules.
  - Engaged in teaching STEM robotics classes using LEGO NXT in STEM club at Grammar Newton International Bilingual School, Hanoi.
  - **Supervisor:** Mr. Nguyen Quoc Hung (Email)

Working Experience

ETEK Automation Solutions JSC Hanoi, Vietnam
Intern *03/2025 - 05/2025*

- **Project:** AGV-based Virtual Simulation for Automated Warehouse Logistics – VINFAST Hai Phong.
  - Sketched AGV/AMR traffic intersections using AutoCAD and simulated their operations using Visual Components Premium 4.6.
  - Monitored and analyzed AGV/AMR behavior, identified errors, and troubleshooting.
  - **Supervisor:** Mr. Nguyen Ngoc Khanh (Email)

Vietnam E-Commerce and Digital Economy Agency, Ministry of Industry and Trade Hanoi, Vietnam
Collaborator *07/2023 - 09/2023*

- **Project:** Program for Consultancy Support and Sales Policy Development for E-Commerce Website (CPAPER)
  - Coordinated with SMEs across Dak Nong and An Giang province to develop e-commerce platforms.
  - Assisted in website design, content management, and prepared technical mannual documents.

Publication

1. **Long Nguyen Khac**. et al. *Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Robot.* International Conference on Advances in Information and Communication Technology, doi: 10.1007/978-3-031-80943-9_57.
   - *Main highlights: an adaptive fuzzy logic control system integrated with backstepping control for a 3-DOF delta robot.*
2. **Long Nguyen Khac**. et al. *YOLOv5s Model with Applied Activation Functions for Personal Protective Equipment Detection in Construction Sites.* Proceedings of National Conference on Smart Technology Applications in Industry 4.0 Smart City, and Sustainability. (2023)
   - *Main highlights: a deep learning-based approach using YOLOv5s with 06 activation functions with a new dataset of 2632 images for detecting PPE.*

Talks

The 3rd International Conference ICTA 2024 — Phu Tho, Vietnam
Mechatronics and Automation — *11/2024*

Honors and Awards

- Selected as one of 20 VNU students received **DiscoverNUS & NUS ASEAN Master's Scholarship** for one semester exchange fully funded by National University of Singapore. (Semester 1, AY25/26)
- Selected as one of 28 global students received **Taiwan Experience Education Program (TEEP) Scholarship** fully-funded by Ministry of Education, Republic of China (Taiwan) to intern at National Chung Cheng University (CCU).
- Selected as one of 04 VNU students to participate in **ASEAN in Today's World** in Indonesia funded by Kyushu Univeristy, Japan.(2025)
- Selected as one of 20 undergraduate and graduate students received **TOSHIBA scholarship**. (AY 24/25)
- Received **Academic Encouragement Scholarship**, 4/6 semesters.(Semester 1, AY22/23, Semester 1, AY 23/24, Semester 1,2 AY 24/25)
- Selected as one of 20 VNU students to participate in **Temasek Foundation Specialist's Community Action & Leadership Exchange Program** in Singapore fully funded by Temasek Foundation. (2024)
- Received **Outstanding Young Face** Award, VNUIS. (2024)
- Received **Student with Five Good Merits** Award, VNU. (2024)
- Received **Fighting Spirit Prize** in South-East Asia Circuit and System Society Hackathon. (2022)

Skills

General skills: Mathematical modeling, control systems designing, embedded programming, mechanical designing
Programming Skills: Matlab, Python, Embedded C/C++
Specialisation: Adaptive Control, Machine Learning, Robotics
Softwares and Frameworks: Matlab/Simulink, Solidworks, PyTorch
Languages: Vietnamese (Native), English (IELTS 6.5: Writing 7.0), Chinese (Beginner)

References

- **Prof. Her-Terng Yau**
  Director of AIM-HI, National Chung Cheng University, Taiwan
  Email: yauherterng@gmail.com
- **Assoc. Prof. Nguyen Nhu Tung**
  Senior Lecturer, Faculty of Engineering and Technology
  Academic Advisor of Industrial Systems Engineering and Logistics Major
  Email: tungnn@vnu.edu.vn
- **Dr. Le Xuan Hai**
  Academic Advisor of Automation and Informatics Major
  Director, Intelligent Control and Artificial Intelligence Lab (ICL)
  Vice Dean, Faculty of Engineering and Technology, International School, VNU
  Email: hailx@vnu.edu.vn